

POWERED BY DATABRICKS + GENIE

Campus Twin

TEAM PAANCH DOST

Tanmay Sachan
Utkarsh Maheshwari
Aditya Kumar Singh
Jawin Roys Fernandes
Saksham Pandey

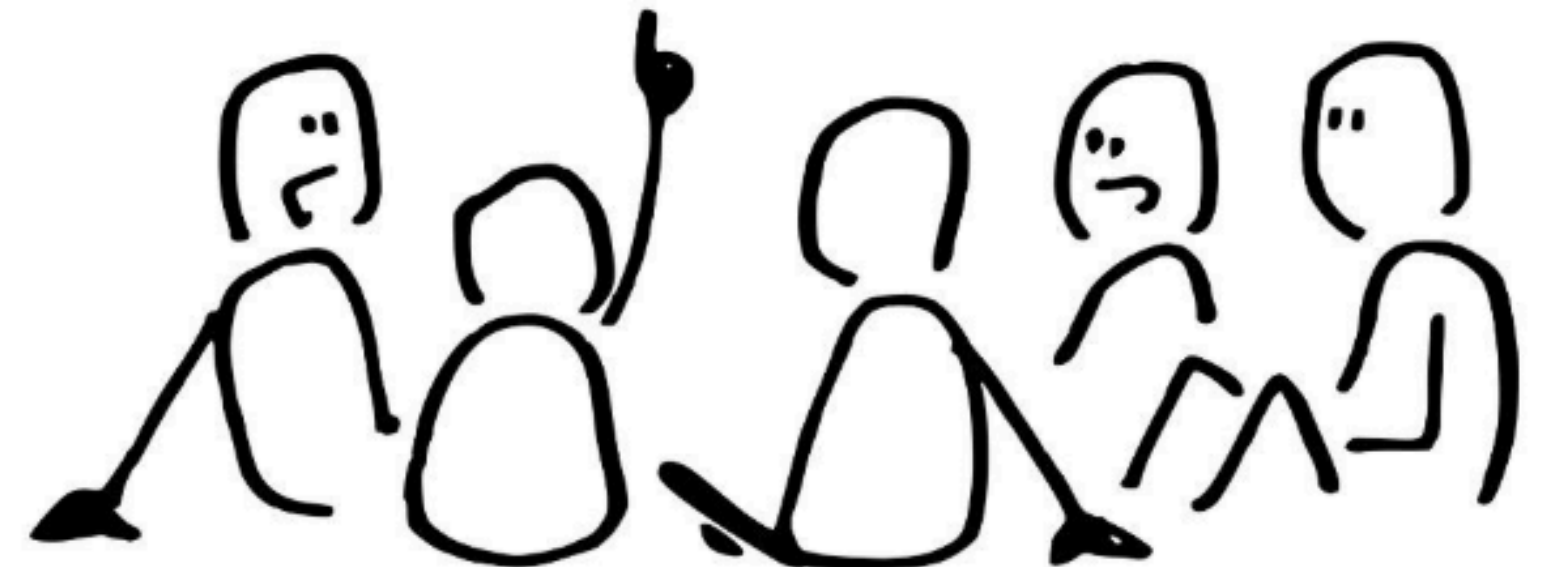
About Us

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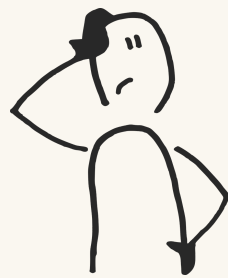
We are a team of five fourth-year engineering students bringing together strong technical skills, real-world industry exposure, and a shared interest in solving problems that matter on campus.

Our team has gained hands-on experience through internships at companies including HPE, First Citizen, Future Standard, and others, giving us exposure to building, deploying, and working with real-world technology.

We believe the next generation of campus infrastructure should be data-driven, intelligent, and experimental. CampusTwin is our attempt to turn that vision into a practical system where decisions can be tested digitally before being implemented in the real world.



Project Overview



Problem

Campus data is everywhere, but campus intelligence is fragmented. Timetables, classrooms, labs, transport, energy, events and student data often exist across disconnected systems. Administrators can see individual metrics, but struggle to understand how one decision affects the rest of the campus.



Solution

A living digital twin of the campus. CampusTwin brings academic, infrastructure and operational data into one unified model and lets users interact with it through Databricks Genie. Users can ask questions, investigate problems and simulate changes such as reallocating classrooms, moving students, adjusting transport or changing facility schedules.



Why It Matters

Don't experiment on the real campus first. CampusTwin lets administrators test decisions in a digital environment and see their predicted impact before implementing them in reality. A single change can affect capacity, utilization, energy, scheduling, transport and resources. CampusTwin helps uncover these cascading effects and identify better decisions.

INTRODUCTION

Project Objective

Main Goal

To create a living digital twin of a college campus that enables data-driven decisions through natural-language intelligence and what-if simulation.

Instead of simply monitoring campus operations, CampusTwin will allow administrators to understand the current campus, predict future conditions, and test proposed changes digitally before implementing them in the real world.

Supporting Objectives

- Unify campus data
- Enable natural-language interaction
- Simulate campus decisions
- Identify cascading effects
- Recommend better decisions



What We Found

1

CAMPUS DATA IS FRAGMENTED

Timetables, room bookings, lab usage, energy, transport and event information are often maintained across different systems and formats.

2

CAMPUS SYSTEMS ARE CONNECTED

A change in one area can affect others.

3

DECISIONS ARE HARD TO TEST

Administrators can analyze historical data, but there is rarely a way to safely ask: "What happens if we change this?" before making the change in the real campus.

Key Insight

Campus doesn't need more dashboards. It needs a way to experiment with decisions.

Who We Designed For



Target User

Campus Administrators & Operations Teams: People responsible for classrooms, laboratories, facilities, transport, energy, scheduling and resource allocation.



Their Challenges

- Data is scattered
- Systems don't connect
- Usage is reactive
- Changes aren't tested
- Answers take manual effort

Work Orders Overview

Open Work Orders

124

In Progress

78

Completed

325

Maintenance Requests

New Requests

28

Urgent Issues

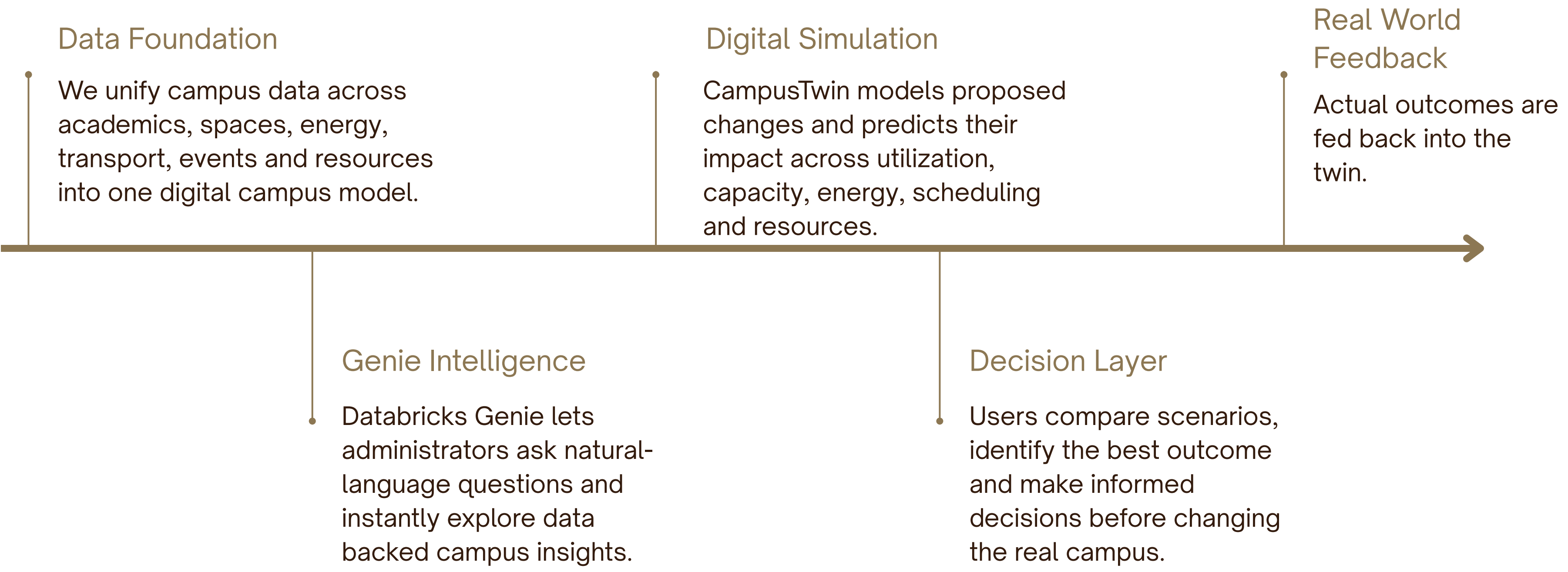
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Work Order Status

Category Breakdown

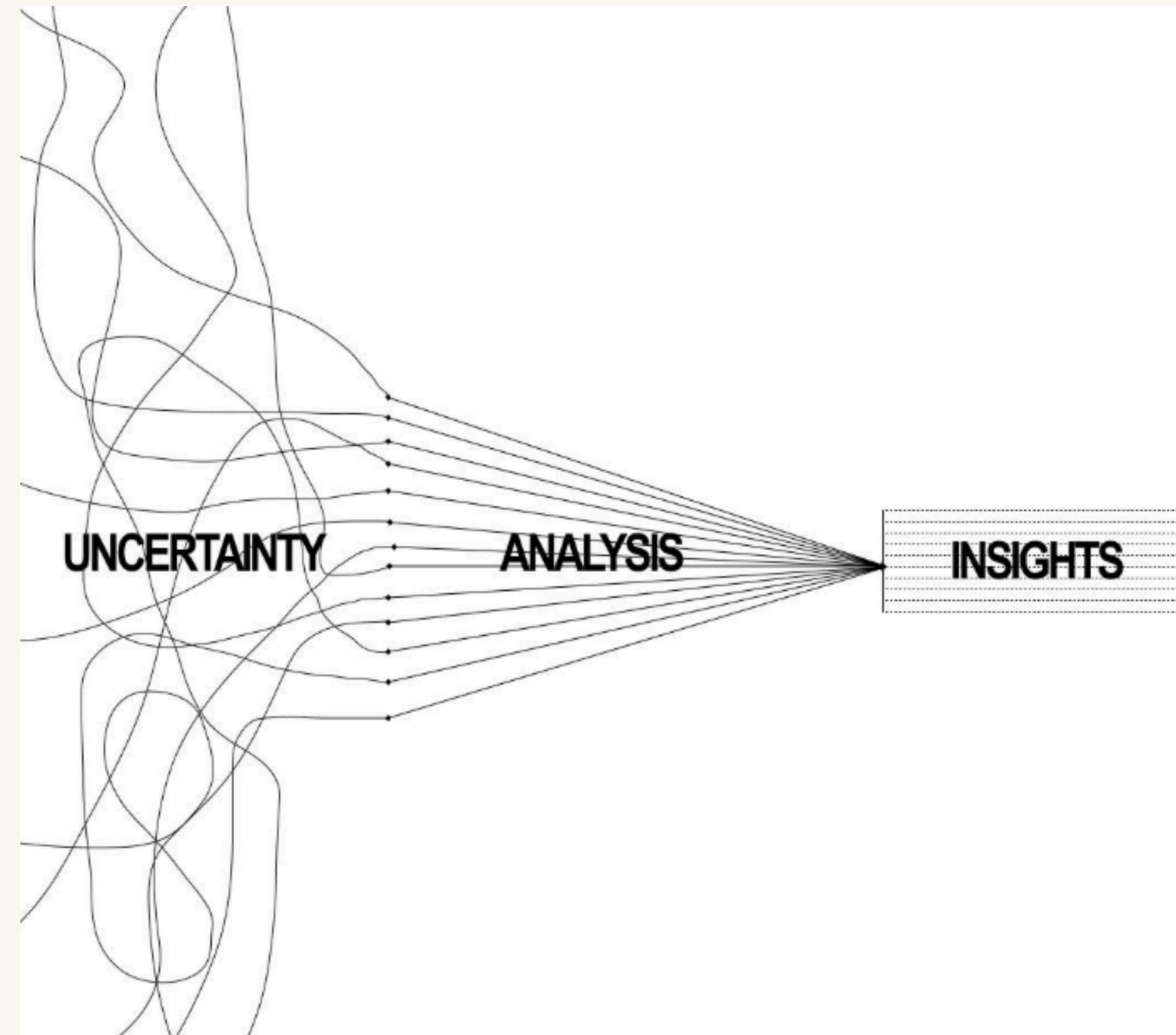
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Solution



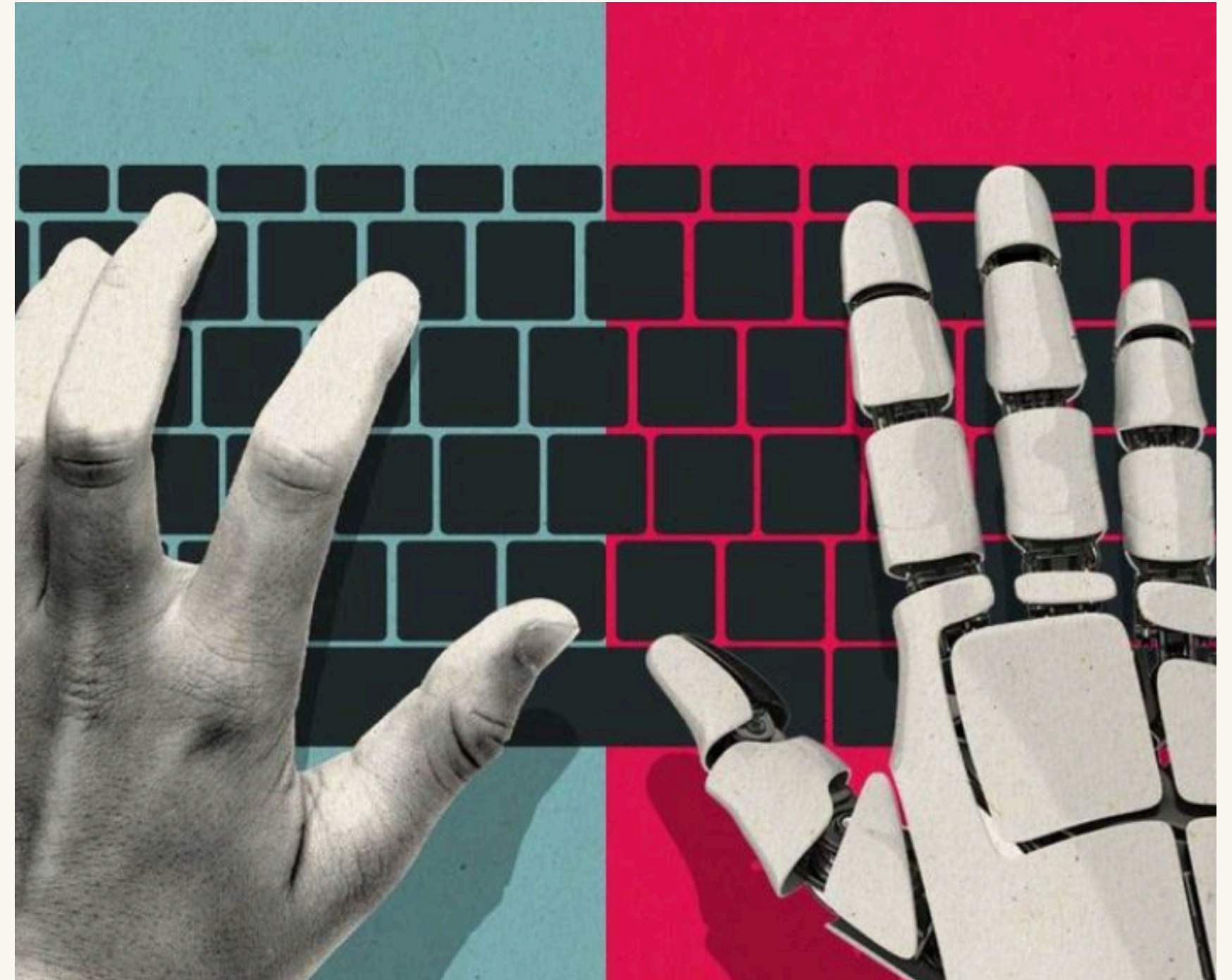
Data Foundation

Campus Twin begins by creating a unified data foundation for the entire campus. Data from classrooms, laboratories, buildings, student enrollment, timetables, transportation, events, energy consumption, and resource usage is collected from different sources and standardized into a common structure using Databricks. Delta Lake provides the central, reliable storage layer, while data pipelines continuously process new and updated information. This creates a single source of truth that represents the current operational state of the campus and provides the historical data required for analysis and prediction.



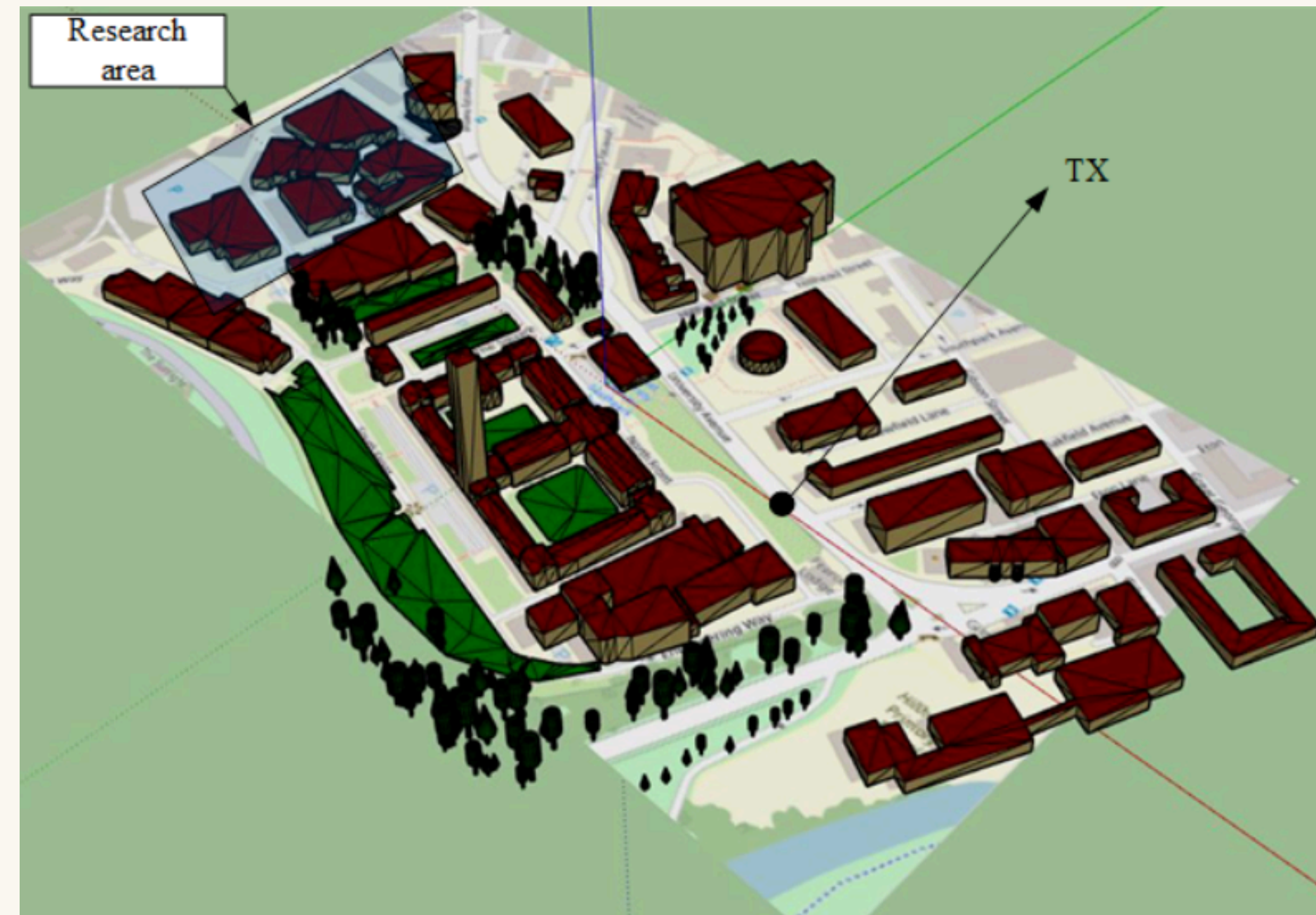
Genie Intelligence

Databricks Genie becomes the natural-language intelligence layer through which administrators interact with the digital twin. Instead of manually searching through spreadsheets, dashboards, or databases, users can simply ask questions such as "Which laboratories are underutilized?", "Why is this building consuming more energy?", or "Which classrooms are available tomorrow?". Genie interprets the question, accesses the relevant campus datasets in Databricks, performs the required analysis, and returns a clear, data-backed answer with relevant metrics and visualizations. This makes complex campus data accessible to users without requiring them to understand SQL or the underlying data structure.



Digital Simulation

The digital twin maintains a virtual representation of the campus and its interconnected resources, allowing proposed changes to be tested before they are implemented physically. A user can propose a scenario such as moving students between laboratories, changing classroom allocations, increasing student intake, modifying bus schedules, or temporarily closing a facility. CampusTwin applies the proposed change to the digital model and evaluates its predicted effects on utilization, capacity, energy consumption, scheduling conflicts, transportation demand, and other dependent resources. This allows administrators to experiment with different scenarios without disrupting the real campus.



Decision Layer

Once a scenario is simulated, Campus Twin turns the results into an actionable decision. Multiple possible solutions can be compared based on factors such as resource utilization, cost, capacity, energy consumption, scheduling conflicts, and operational constraints. Instead of simply showing numbers, the system highlights the trade-offs between alternatives and identifies which option produces the most favorable overall outcome. Genie can then explain why a particular option is recommended, allowing administrators to move from a natural-language question to an evidence-based decision backed by the campus data and simulations.



Real-World Feedback

CampusTwin continues to evolve after a decision is implemented by comparing predicted outcomes with what actually happens on campus. For example, if the twin predicts that moving a class will increase laboratory utilization to 78%, the actual utilization after implementation can be measured and compared against that prediction. These differences become feedback for improving future analysis and predictive models. Over time, the system develops a more accurate understanding of the specific campus, allowing its simulations and recommendations to become increasingly reliable. This creates a continuous loop of simulate, implement, observe, learn, and improve.



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The End

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